HUMAN-AI INTERACTION I

Game Assistant

Group Y

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Project Topic

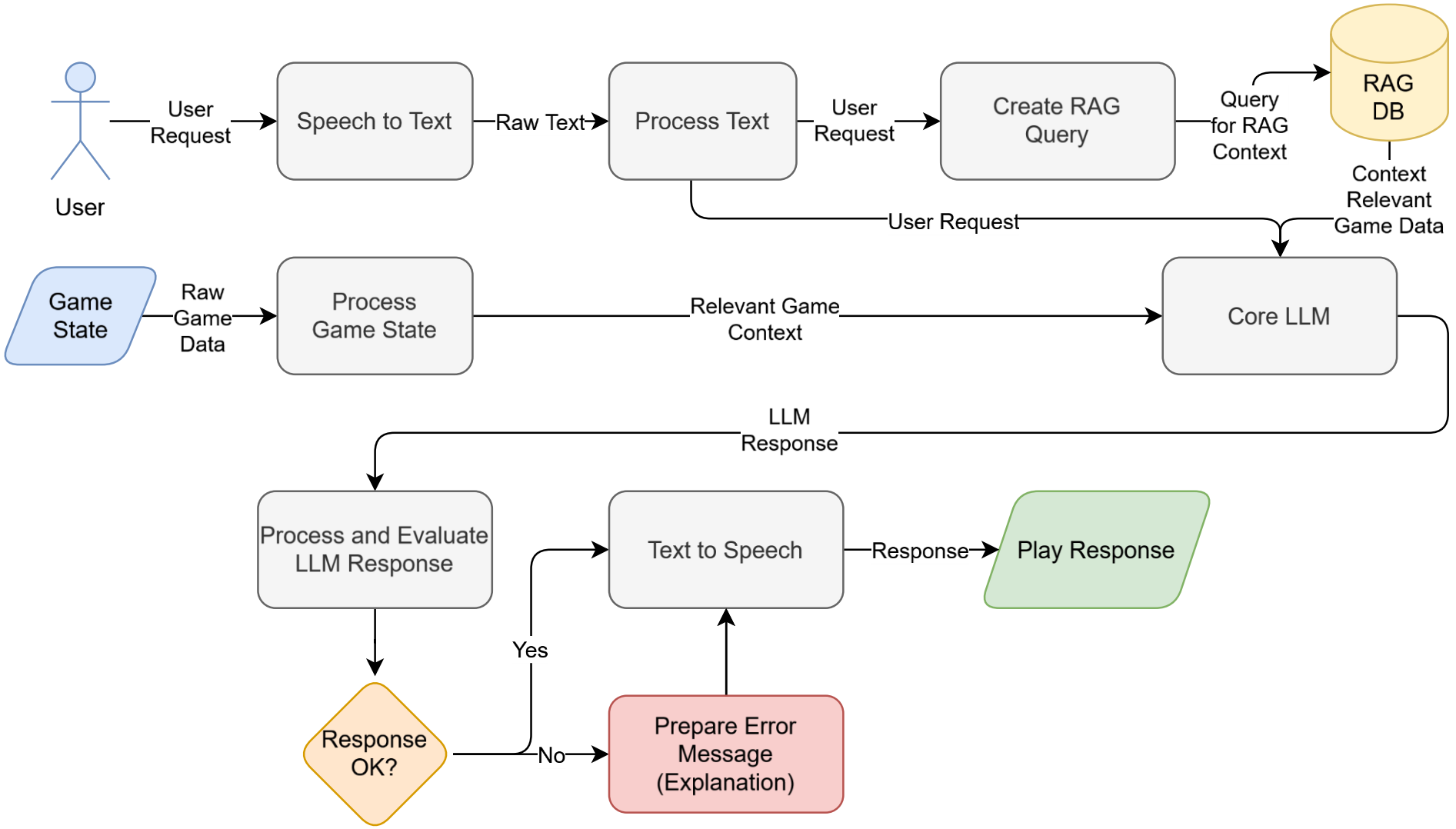
- Playing a new game can be challenging. The players are often presented with too much content too quickly.
- We want to help players through their journey by guiding them whenever they request it.
- The Game Assistant is a Speech-to-Speech AI application that can answer the player any game-related questions they have while playing.

AI In your Project

AI in your project - Overview

- Multiple AI Components
  - STT (Speech to Text) - transcribing input
  - LLM (Text to Text) - processing user prompt*
  - TTS (Text to Speech) - generating output
- Input/Perception: User's speech (audio)
- Output/Action: Response as speech (audio)

AI in your project - Flow



AI in your project - Reasoning

- Current scope: Food and cooking in Minecraft
- Player: "How do I get bread?"
- Player inventory (from Game State): [2x wheat, (…)]
- RAG query for item "Bread" fetches context info
- User wants "Bread", bread is made from (3x) wheat (…)
- Answer: "You can craft bread from three wheat. You have two wheat so you need one more wheat to craft bread!"

AI in your project - Technicals

- STT Model: OpenAI Whisper (faster-whisper, whisper-tiny)
- LLM: Ollama, Mystral, Qwen3 (instruction tuned)
- TTS: Piper, Deepgram (API)
- Optional: LM for internal prompt and game state processing?

AI in your project - Data

- Data/Knowledge-Base for RAG
  - General game data
  - Preprocessed and stored locally
- Memory for the System? (remember the last N user prompts?)
- Remember user preferences for the session?
- No session data is shared between instances

AI in your project - Understanding

- AI needs to understand what item is referenced in the game context
  - Filter/process the player prompt for keywords
  - Ensure keywords are valid (context in scope)

- AI must deliver an accurate, reasonable and understandable explanation

- We must account for and correctly handle
  - Out-of-context questions
  - Missing context (from RAG)
  - Faulty responses (i.e. hallucinations)

AI in your project - Convention

- Agent needs to know about
  - it's application (game assistant)
  - game specifics (type of game, mechanics)

- Agent may adapts to the player by knowing the
  - player preferences (efficiency vs exploration)
  - player type (newbie, recurring player)

Human In your Project

Human Interface in your project – Input Output Interface

- Player input via microphone
- Player gets output via speakers
- GUI to visualize input and output (history)
- Store interaction history to a log file

Human Interface in your project – Understanding the AI

- Output must not be too short or too long
- On point
- Easy to understand
- Explainable

Human Interface in your project - Technologies

- What technologies will you use in the interface for human input?
- Interface technologies: Whisper, Piper
- User feedback included in the response (error message)
- Data storage: Conversation history in GUI, stored user preferences

The Plan

Roadmap

- M1 December 15. 2025
  - Finished S2S pipeline
  - Dummy input is successfully processed

- M2 January 12. 2026
  - Finished prototype

- Requirements or needs:
  - Do you need access to hardware? - No
  - Do you need access to software? - No
  - Will you work in the lab? - No

Thank you!

Next class